

Project Workflow Diagram & Pipeline Details

Detailed Stages of the Data Engineering & ML Pipeline

Stage 1: Data Collection & Ingestion

The pipeline ingests two datasets: `application_record.csv` (demographics of 5,000 applicants) and `credit_record.csv` (170,000+ monthly payment status history records). The files are merged using the applicant's unique identifier (ID).

Stage 2: Exploratory Data Analysis (EDA)

Visualizations examine the features individually (univariate analysis of income, age, education, and family sizes) and examine their joint distributions (bivariate scatters and correlation heatmaps).

Stage 3: Data Preprocessing & Target Engineering

- **Duplicates Check:** Drop duplicate records to prevent training model bias.
- **Missing Values:** Impute missing occupations with 'Unknown'.
- **Date Normalization:** Convert negative day counters into positive years (e.g. Age and Employed Years).
- **Target Variable (y) Creation:** Delinquencies in monthly credit status tracking are aggregated: if an applicant's maximum overdue status ever exceeds or equals 1 (30+ days overdue), the target label is high-risk ($y=1$); otherwise, it is low-risk ($y=0$).
- **Categorical Handling & Scaling:** Apply `LabelEncoder` to text categoricals and `StandardScaler` to normalize all numerical predictors.

Stage 4: Model Training, Selection & Deployment

Compare Logistic Regression, Decision Tree, and Random Forest models using stratified splits. Automatically select the best-performing estimator, serialize it, and integrate it into the Flask web server for local browser execution.

Workflow Pipeline Flowchart

Project Development Workflow

